

Algorithms

Lecture #19

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Design and Analysis of Algorithms

Analysis: Edit Distance

Analysis: Edit Distance

- There are $\Theta(n^2)$ entries in the matrix.
- Each entry $E(i, j)$ takes $\Theta(1)$ time to compute.
- The total running time is $\Theta(n^2)$.

Chain Matrix Multiply

Chain Matrix Multiply

- Suppose we wish to multiply a series of matrices:

$$A_1 A_2 \dots A_n$$

- In what order should the multiplication be done?

Matrix Multiplication

Matrix Multiplication

- A $p \times q$ matrix A can be multiplied with a $q \times r$ matrix B.
- The result will be a $p \times r$ matrix C.
- In particular, for $1 \leq i \leq p$ and $1 \leq j \leq r$,

$$C[i, j] = \sum_{k=1}^q A[i, k]B[k, j]$$

Matrix Multiplication

Matrix Multiplication

- There are $(p \cdot r)$ total entries in C and each takes $O(q)$ to compute.

$$C[i, j] = \sum_{k=1}^q A[i, k]B[k, j]$$

- Thus the total number of multiplications is $p \cdot q \cdot r$.

Chain Matrix Multiplication

Chain Matrix Multiplication

- Consider the case of 3 matrices:
 A_1 is 5×4 , A_2 is 4×6 and A_3 is 6×2
- The multiplication can be carried out either as
 $((A_1A_2)A_3)$ or $(A_1(A_2A_3))$.
- The cost of the two is
 $((A_1A_2)A_3) = (5 \cdot 4 \cdot 6) + (5 \cdot 6 \cdot 2) = 180$
 $(A_1(A_2A_3)) = (4 \cdot 6 \cdot 2) + (5 \cdot 4 \cdot 2) = 88$

Chain Matrix Multiplication

Chain Matrix Multiplication

- There is considerable savings achieved even for this simple example.
- In general, however, in what order should we multiply a series of matrices $A_1 A_2 \dots A_n$.
- Matrix multiplication is an associative but not commutative operation.

Chain Matrix Multiplication

Chain Matrix Multiplication

- In general, however, in what order should we multiply a series of matrices $A_1 A_2 \dots A_n$.
- Matrix multiplication is an associative but not commutative operation.
- We are free to add parenthesis the above multiplication but the order of matrices can not be changed.

Chain Matrix Multiplication

Chain Matrix Multiplication

Chain Matrix Multiplication Problem

Given a sequence $A_1, A_2, \dots, A_n$ and dimensions $p_0, p_1, \dots, p_n$ where A_i is of dimension $p_{i-1} \times p_i$, determine the order of multiplication that minimizes the number of operations.

Chain Matrix Multiplication

Chain Matrix Multiplication

- We could write a procedure that tries all possible parenthesizations.
- Unfortunately, the number of ways of parenthesizing an expression is very large.

Chain Matrix Multiplication

Chain Matrix Multiplication

- If there are n items, there are $n - 1$ ways in which outer most pair of parentheses can placed.

$(A_1)(A_2 A_3 A_4 \dots A_n)$
or $(A_1 A_2)(A_3 A_4 \dots A_n)$
or $(A_1 A_2 A_3)(A_4 \dots A_n)$
.....
or $(A_1 A_2 A_3 A_4 \dots A_{n-1})(A_n)$

Chain Matrix Multiplication

Chain Matrix Multiplication

- Once we split just after the k th matrix, we create two sublists to be parenthesized, one with k and other with $n - k$ matrices.

$$(A_1 A_2 \dots A_k) (A_{k+1} \dots A_n)$$

- We could consider all the ways of parenthesizing these two.

Chain Matrix Multiplication

Chain Matrix Multiplication

- Since these are independent choices, if there are L ways of parenthesizing the left sublist and R ways to parenthesize the right sublist, then the total is $L \cdot R$.

Chain Matrix Multiplication

Chain Matrix Multiplication

- This suggests the following recurrence for $P(n)$, the number of different ways of parenthesizing n items:

$$P(n) = \begin{cases} 1 & \text{if } n = 1, \\ \sum_{k=1}^{n-1} P(k)P(n-k) & \text{if } n \geq 2 \end{cases}$$

Chain Matrix Multiplication

Chain Matrix Multiplication

- This is related to a famous function in combinatorics called the Catalan numbers.
- Catalan numbers are related the number of different binary trees on n nodes.
- Catalan number is given by the formula:

$$C(n) = \frac{1}{n + 1} \binom{2n}{n}$$

Chain Matrix Multiplication

Chain Matrix Multiplication

- In particular, $P(n) = C(n - 1)$
- $C(n) \in \Omega(4^n/n^{3/2})$
- The dominating term is the exponential 4^n thus $P(n)$ will grow large very quickly. So this approach is not practical.

Chain Matrix Multiplication-DP

Chain Matrix Multiplication-Dp

- The dynamic programming solution involves breaking up the problem into subproblems whose solutions can be combined to solve the global problem.
- Let $A_{i..j}$ be the result of multiplying matrices i through j .

Chain Matrix Multiplication-DP

Chain Matrix Multiplication-Dp

- Let $A_{i..j}$ be the result of multiplying matrices i through j .
- It is easy to see that $A_{i..j}$ is a $p_{i-1} \times p_j$ matrix.

$$\begin{array}{cccccc} A_3 & A_4 & A_5 & A_6 & = & A_{3..6} \\ 4 \times 5 & 5 \times 2 & 2 \times 8 & 8 \times 7 & & 4 \times 7 \end{array}$$

Chain Matrix Multiplication-DP

Chain Matrix Multiplication-Dp

- At the highest level of parenthesization we multiply two matrices
$$A_{1..n} = A_{1..k} A_{k+1..n} \quad 1 \leq k \leq n-1.$$
- The question now is: what is the optimum value of k for the split and how do we parenthesis the sub-chains $A_{1..k}$ and $A_{k+1..n}$.

Chain Matrix Multiplication-DP

Chain Matrix Multiplication-Dp

- We can not use divide and conquer because we do not know what is the optimum k .
- We will have to consider all possible values of k and take the best of them.
- We will apply this strategy to solve the subproblems optimally.

Chain Matrix Multiplication

Chain Matrix Multiplication

Dynamic Programming Formulation

- We will store the solutions to the subproblem in a table and build the table bottom-up (why)?
- For $1 \leq i \leq j \leq n$, let $m[i, j]$ denote the minimum number of multiplications needed to compute $A_{i..j}$.

Chain Matrix Multiplication

Chain Matrix Multiplication

- For $1 \leq i \leq j \leq n$, let $m[i, j]$ denote the minimum number of multiplications needed to compute $A_i..j$.
- The optimum can be described by the following recursive formulation.

Chain Matrix Multiplication